

Week 12 Progress Update: Plan for Completion

Project: RoboCup HSL 2026, Booster K1 Localisation

Team: A3-04-HSL

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Branch: PAR/develop

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1. Project Overview

We are developing and evaluating a particle filter based localisation system for the Booster K1 humanoid robot competing in RoboCup HSL 2026. The project addresses two core challenges in RoboCup localisation:

1. Robust localisation under uncertainty: a particle filter replaces the existing `MultiModalCMKF` to provide pose estimation with ESS-based adaptive resampling, directional/directionless observation models, and observation-driven seeding for kidnapped-robot recovery.
2. Heading ambiguity on a symmetric field: a visual natural landmark system using ORB feature detection helps resolve orientation ambiguity caused by the RoboCup field's bilateral symmetry.

A complementary localisation replay infrastructure, powered by HTC Vive motion tracking, provides external ground-truth pose for offline parameter optimisation and rigorous evaluation of both systems.

2. Particle Filter (Tom & Mark)

2a. Minimum Viable Product

The particle filter (`ParticleFilter.cpp`) has been fully implemented as a single-hypothesis localiser that replaces the existing `MultiModalCMKF`. The MVP includes:

Feature	Status	Details
Motion model (odometry)	Complete	Shifts particles by <code>OdometryDiff</code> with configurable noise
Observation model (directional)	Complete	3-DOF Mahalanobis distance for corners, T-junctions, X-junctions, centre circle, goal posts
Observation model (directionless)	Complete	1-DOF Gaussian on radial distance for penalty spots, centre circle directionless, YOLO features
ESS-based adaptive resampling	Complete	Triggers multinomial resampling when effective sample size < threshold
Observation-driven seeding (PF-16)	Complete	Injects Gaussian clusters when uncertainty > 800 mm
State transition handling	Complete	Game state resets, penalties, manual placement, etc.
Roughening after resampling	Complete	Gaussian jitter to prevent particle impoverishment
Config file loading	Complete	<code>ParticleFilterParams.cfg</code> with <code>boost::program_options</code>
Informed particle initialisation with directional features	Complete	Spawn particles at possible positions relative to an observed directional feature

2b. Plan for Completion

The remaining work is structured into four work packages. The first three are core deliverables; the fourth (IMU) is a bonus/stretch goal.

1. Test and evaluate uncertainty metrics for robot pose such as **covariance matrix** or **determinant**
 2. Implement **informed initialization of particles** based on **directionless features**
 3. **Cull off-field particles** to help reduce flipping and redundant computation
 4. Add **IMU as an input feature for the particle filter** (bonus for AS3)
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3. Localisation Replay & External Pose Estimation (Sam)

3a. Minimum Viable Product

Sam has implemented the localisation replay infrastructure on branch `par/localisation_replay`:

Feature	Status	Details
HTC Vive external pose estimation	Complete	ROS2 integration with Vive trackers and base stations
Localisation replay recorder (<code>LocaliserReplayRecorder</code>)	Complete	Records estimator inputs (odometry, field features) and outputs
Localisation replay runner (<code>LocaliserReplayRunner</code>)	Complete	Replays recorded data through the localiser offline
External vs internal pose comparison	Complete	Compares Vive ground truth with filter estimate

3b. Plan for Completion

#	Task	Details
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5.1	Record experiment: untuned PF using external pose	Run robot recording Vive ground truth + PF output simultaneously
5.2	Implement automated parameter optimiser	Script searching parameter space to minimise position error vs Vive ground truth
5.3	Redo experiment with tuned filter	Record second run using optimised parameters
5.4	Document optimisation results	Compare pre-tuning vs post-tuning error statistics

4. Natural Landmarks (Eric)

The project aims to reduce robot localisation failures caused by orientation flipping in a symmetric RoboCup soccer field. To address this issue, a visual landmark recognition system based on ORB features is being developed and integrated with the existing localisation framework.

4a. Minimum Viable Product

The following components have been completed:

- Implementation of ORB feature detection and descriptor extraction.
- Development of a persistent keypoint storage system to maintain visual landmarks across frames.
- Implementation of keypoint matching between current observations and stored landmarks.
- Initial testing of keypoint persistence and matching performance in a soccer field

Preliminary testing indicates that the system can successfully identify recurring visual features and maintain a set of persistent landmarks over time. The current work has focused on establishing a reliable visual feature pipeline before integration with the robot's localisation system.

4b. Plan for Completion

The remaining work is focused on incorporating visual landmark information into the robot's existing state estimation framework. The proposed timeline is:

Week 1

- Refine keypoint selection and storage criteria to improve landmark stability.
- Implement confidence scoring for matched landmarks.
- Evaluate match consistency under different robot orientations and viewpoints.

Week 2

- Integrate visual landmark observations into the existing localisation filter (Kalman Filter or Particle Filter).
- Develop a measurement model that converts landmark matches into a heading confidence estimate.
- Use this information to bias the localisation solution away from incorrect symmetric orientations.

Week 3

- Conduct systematic testing and further parameter tuning.
- Analyse failure cases and improve robustness.
- Prepare demonstration scenarios and experimental results.

4c. Improvements Beyond the Minimum Viable Solution

Several improvements are planned beyond this baseline:

- Use match confidence scores instead of only counting the number of matches.
- Use observations from multiple frames to reduce the effect of incorrect matches.
- Prioritise visual landmarks that remain stable over time.
- Check that matched landmarks are geometrically consistent to remove false matches.
- Keypoint storing should only be active when the robot is confident in where it is

These improvements aim to increase robustness and reduce false heading corrections during gameplay.

4d. Proposed Experimental Evaluation

The evaluation will focus on three key tests: flip resolution accuracy, reduction in localisation flips, and qualitative recovery from a flipped state. These tests are chosen because they directly

measure whether the ORB-based persistent keypoint system improves the robot's ability to resolve heading ambiguity on a symmetric soccer field.

Test 1: Flip Resolution Accuracy

Purpose

This test measures how often the robot correctly identifies its true heading when the localisation system has two possible symmetric orientation estimates.

Setup

The robot will have time to collect keypoints then get its vision taken away. It will be walked to several predefined locations on the soccer field where the visual appearance or field geometry may cause localisation ambiguity. At each location, the robot will be manually oriented toward one of the two symmetric directions.

The test will be run in two modes:

1. Baseline localisation only.
2. Localisation with ORB persistent keypoint assistance.

Procedure

1. Give the robot time to store keypoints
2. Take away its vision
3. Walk the robot to a selected test position.
4. Observe the robot's true heading.
5. Allow the localisation system to estimate the heading.
6. Record whether the estimated heading matches the true heading.
7. Repeat the test for both symmetric headings.
8. Repeat across multiple field positions.

Number of Trials

- 3 field positions.
- 2 heading directions per position.
- 3 repeated trials per direction.

Data to Record

For each trial, record:

- Test position.
- True heading.
- Estimated heading.
- Whether the estimate was correct.
- Whether the system reported uncertainty.
- Number of persistent keypoints.
- Number of accepted matches.
- ORB heading confidence score.

Success Criteria

The ORB-assisted system should correctly identify the true heading more often than the baseline system.

Metrics

Flip resolution accuracy = correct heading decisions / total trials

A successful outcome would be a clear improvement in correct heading decisions compared with the baseline.

Test 2: Reduction in Localisation Flips

Purpose

This test measures whether the ORB-based system reduces the number of times the robot flips to the incorrect symmetric orientation during operation.

Setup

The robot will operate on the soccer field under controlled but realistic movement conditions. The same movement pattern should be used for both baseline and ORB-assisted testing.

The test will be run in two modes:

1. Baseline localisation only.
2. Localisation with ORB persistent keypoint assistance.

Procedure

1. Start the robot from a known position and heading.
2. Run the robot for a fixed time period.
3. Use the same movement pattern for each trial, such as:
 - walking,
 - rotating in place,
 - Rotating while walking.
4. Record the localisation estimate throughout the run.
5. Count every time the heading estimate flips to the incorrect symmetric orientation.
6. Record how long the robot remains in the incorrect orientation before recovering.
7. Repeat the test several times for both baseline and ORB-assisted modes.

Number of Trials

- 3 baseline runs.
- 3 ORB-assisted runs.
- Each run lasting 3–5 minutes.

Data to Record

For each run, record:

- Test mode: baseline or ORB-assisted.
- Trial duration.
- Number of localisation flips.
- Average flips per minute.
- Total time spent in the incorrect orientation.
- Average recovery time after a flip.
- Number of ORB matches over time.
- Confidence score over time.

Metrics

Flips per minute = number of flips / trial duration

Incorrect orientation time = total time robot spends facing the wrong symmetric heading

Average recovery time = total recovery time / number of flips

Success Criteria

The ORB-assisted system should show:

- fewer localisation flips,
- shorter time spent in the incorrect orientation,
- faster recovery after a flip,
- more stable heading estimates over time.

Even if the system does not eliminate all flips, a measurable reduction compared with the baseline would demonstrate project success.